

Critical Gevrey Derivative Loss: Exact Later Answer to Open Problem 2

Run fa_banach_001, agent_lane_14

11 August 2026

Status and sources

Status: literature_already_answered (Open Problem 2 only).

The original question is Open Problem 2 on PDF page 24 of Marina Ghisi and Massimo Gobino's *Residual pathologies*, arXiv:1908.09496 [1]. Their distinct later paper *Critical counterexamples for linear wave equations with time-dependent propagation speed*, arXiv:1909.10020 [2], answers it in Theorem 3.2 on PDF page 11; the relevant initial-data construction is on PDF pages 23–24.

Source question

The source fixes the abstract wave equation with zero initial displacement,

$$u''(t) + c(t)Au(t) = 0, \quad u(0) = 0, \quad u'(0) = \psi,$$

and asks whether there exist

- an α -Hölder speed $c(t)$, $0 < \alpha < 1$,
- a critical-order initial velocity $\psi \in \mathcal{G}_{s,r_0}(A)$ with $s = (1 - \alpha)^{-1}$ and $r_0 > 0$, and
- a time $t_0 > 0$,

such that the corresponding solution has severe derivative loss, or an equivalent form of it, for every $t > t_0$.

Exact later answer

Theorem 3.2 of [2] assumes an unbounded nonnegative discrete multiplication operator A , an α -Hölder strictly positive speed, and

$$s = S = \frac{1}{1 - \alpha}, \quad r_0 > 0.$$

It allows $\delta = 0$ in the more general equation containing a damping term, and sets the explicit finite time

$$t_0 = \frac{32\mu_2^{(1+\alpha)/2}}{H} r_0.$$

It proves that a residual set of admissible propagation speeds admits a solution whose initial data lie in the critical finite-radius Gevrey class and such that

$$(u(t), u'(t)) \notin \mathcal{G}_{-S, \log, 1/2}(A) \times \mathcal{G}_{-S, \log, 0}(A) \quad \text{for every } t > t_0.$$

The implication following the theorem shows that this is the largest loss compatible with the known lower regularity bound.

The proof on PDF page 23 removes the only possible formulation ambiguity: it starts with

$$u(0) = 0, \quad u'(0) = \sum_i a_i e_i,$$

then chooses

$$a_i = \lambda_i^{-1} \exp(-r_0 \lambda_i^{1/s})$$

for Theorem 3.2. Thus the result supplies precisely the initial velocity requested in Open Problem 2, not merely an arbitrary pair of critical initial data.

Why this is an explicit literature answer

The later paper's abstract says it treats critical cases left open by the previous literature. Its introduction identifies the critical $s = (1 - \alpha)^{-1}$ case as the first such case, says Theorem 3.2 proves finite-time severe loss, includes the nondissipative case $\delta = 0$, and cites [1]. The relation is therefore known to the supporting authors and belongs in `literature_already_answered`, not in an agent-generated proof category.

Scope limitations and search record

This packet clears only Open Problem 2. It does not answer Open Problem 1 about approximate differentiability of f' at every point, nor Open Problem 3 about loss of transport regularity in every open subset of the spatial support. Exact-phrase, title, author, and arXiv searches through 11 August 2026 found no later answer to those two questions.

References

- [1] M. Ghisi and M. Gobbino, *Residual pathologies*, arXiv:1908.09496 (2019).
- [2] M. Ghisi and M. Gobbino, *Critical counterexamples for linear wave equations with time-dependent propagation speed*, arXiv:1909.10020 (2019).